

Semantic Understanding and Command Compliance

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1. Course Content

1. After running the large model program, users can interact with the robot through voice dialogue. User voice commands are first converted into text by the speech recognition large model, then the text is used to generate a large model and a visual multimodal system to accurately understand the user's commands and voice. Finally, the robot performs the specified action according to the user's command and responds to the user.

2. Start the Agent

Note: The agent must be started before testing any case. If it is already started, it does not need to be started again.

Enter the following command on the vehicle terminal:

```
start_agent
```

The terminal will print the following information, indicating a successful connection:

```
mircoROS_Agent
[1767693398.286449] info | ProxyClient.cpp | create_datareader | d
subscriber created | client_key: 0x1D229138, subscriber_id: 0x001(4), partici
pant_id: 0x000(1)
[1767693398.290468] info | ProxyClient.cpp | create_topic | t
opic created | client_key: 0x1D229138, topic_id: 0x007(2), participant_
id: 0x000(1)
[1767693398.294835] info | ProxyClient.cpp | create_subscriber | s
subscriber created | client_key: 0x1D229138, subscriber_id: 0x002(4), partici
pant_id: 0x000(1)
[1767693398.299603] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x1D229138, datareader_id: 0x002(6), subscri
ber_id: 0x002(4)
[1767693398.303485] info | ProxyClient.cpp | create_topic | t
opic created | client_key: 0x1D229138, topic_id: 0x008(2), participant_
id: 0x000(1)
[1767693398.330727] info | ProxyClient.cpp | create_subscriber | s
subscriber created | client_key: 0x1D229138, subscriber_id: 0x003(4), partici
pant_id: 0x000(1)
[1767693398.335405] info | ProxyClient.cpp | create_datareader | d
atareader created | client_key: 0x1D229138, datareader_id: 0x003(6), subscri
ber_id: 0x003(4)
```

3. Run the Case

3.1 Start the Program

For Raspberry Pi 5 users, if the ROS container is not running, you need to start it first.

```
bringup_m3
```

Open a terminal inside the container.

```
exec_m3
```

Open the terminal on the vehicle terminal and enter the following command to start:

```
shell ros2 launch m3_bringup llm_agent_control.launch.py
```

Alternatively, you can use the following shortcut:

```
multi_brains
```

After initialization, the following will be displayed:

```
[model_service-2] [INFO] [1767699385.642591925] [ASR_Detect]: paraformer-realtime-v2 ASR_Engine initialized successfully
[model_service-2] [INFO] [1767699385.655452485] [model_service]: Tongyi TTS_Engine initialized successfully
[model_service-2] [INFO] [1767699385.656321749] [model_service]: ROS_LLM_Service Initialization completed
```

The most crucial module logs are:

- action_service: Indicates that the action server started successfully

```
[action_service-3] [INFO] [1767699384.764757965] [action_service]:
ROS_Action_Service Initialization completed
```

- `model_service`: Indicates the corresponding model used by the current speech recognition engine `ASR_Engine` and speech synthesis engine `TTS_Engine`, and also indicates that `model_service` started successfully

```
[model_service-2] [INFO] [1767699385.642591925] [ASR_Detect]: paraformer-
realtime-v2 initialized successfully
[model_service-2] [INFO] [1767699385.655452485] [model_service]: Tongyi
TTS_Engine initialized successfully
[model_service-2] [INFO] [1767699385.656321749] [model_service]: ROS_LLM_Service
Initialization completed
```

3.2 Test Cases

Two reference test cases are provided here. Users can create their own test commands.

- Please move forward quickly 1 meter, then slowly retreat 0.5 meters like a tortoise, then turn 30 degrees to the left, then 90 degrees to the right, then move horizontally 0.5 meters to the left, then horizontally 10 centimeters to the right.
- Please tell a joke.

3.2.1 Case 1

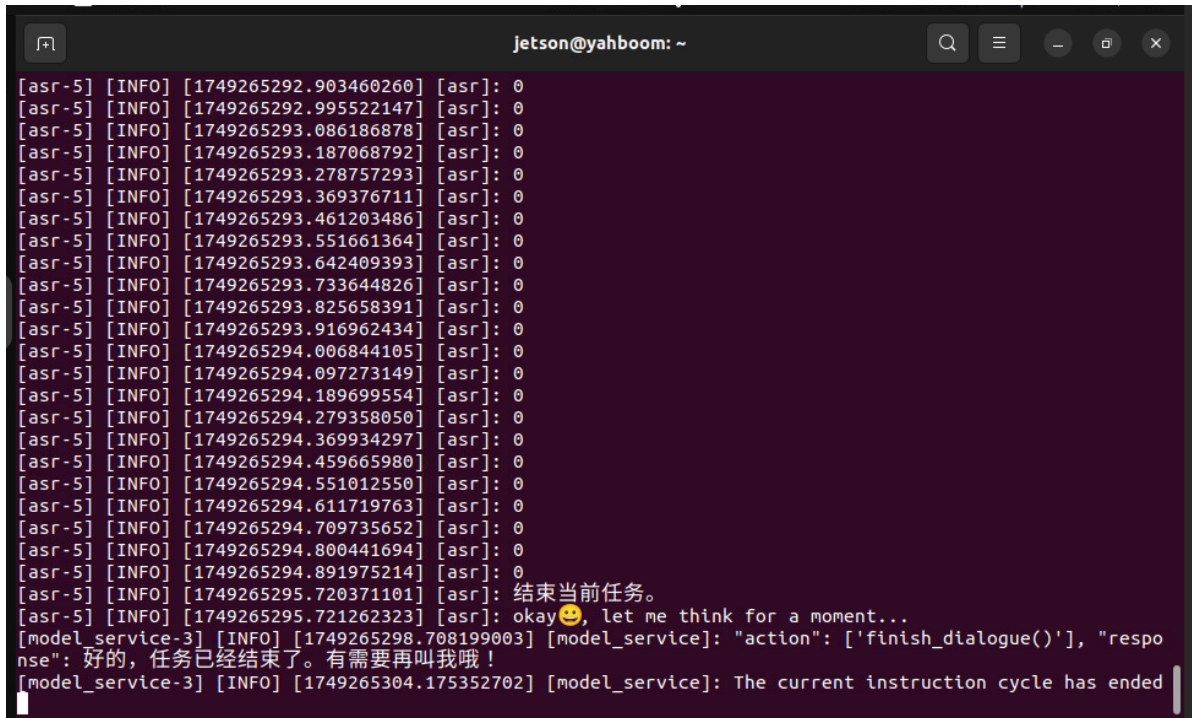
- First, wake up the robot using "Hello yahboom". The robot will respond. After the recording prompt tone, the user can speak. The robot will perform dynamic sound detection. If there is sound activity, it will print "1-1-1-1"; if there is no sound activity, it will print "-----". After speaking, it will perform tail sound detection. If there is silence for more than 1.5 seconds, recording will stop.
- The robot will first respond to the user, and then follow the instructions. The terminal will print the following information:

```
jetson@yahboom: ~
[asr-5] [INFO] [1749262312.221424325] [asr]: 0
[asr-5] [INFO] [1749262312.309843739] [asr]: 0
[asr-5] [INFO] [1749262312.401108892] [asr]: 0
[asr-5] [INFO] [1749262312.491831151] [asr]: 0
[asr-5] [INFO] [1749262312.581573768] [asr]: 0
[asr-5] [INFO] [1749262316.951043317] [asr]: 请你先快速向前前进1米, 然后像乌龟一样缓慢的后退0.5米, 然后
左转30度, 再向右转90度, 再向左平移0.5米, 再向右平移10厘米。
[asr-5] [INFO] [1749262316.952518011] [asr]: okay😊, let me think for a moment...
[model_service-3] [INFO] [1749262322.825880153] [model_service]: The upcoming task to be carried out: 1.
快速前进1米, 缓慢后退0.5米, 左转30度, 右转90度, 向左平移0.5米, 向右平移0.1米。
[model_service-3] [INFO] [1749262331.265122977] [model_service]: "action": ['set_cmdvel(0.5, 0, 0, 2)',
'set_cmdvel(-0.1, 0, 0, 5)', 'move_left(30, 1.5)', 'move_right(90, 1.5)', 'set_cmdvel(0, 0.5, 0, 1)', 's
et_cmdvel(0, -0.1, 0, 1)'], "response": 好的, 我这就开始执行你的指令, 先快速前进再慢慢后退, 然后转来转去
的, 就像在跳一支复杂的舞蹈呢!
[action_service-4] [INFO] [1749262344.531533039] [action_service]: action service execute set_cmdvel(0.
5, 0, 0, 2) succeeded
[action_service-4] [INFO] [1749262350.067616856] [action_service]: action service execute set_cmdvel(-0
.1, 0, 0, 5) succeeded
[action_service-4] [INFO] [1749262350.975963416] [action_service]: action service execute move_left(30,
1.5) succeeded
[action_service-4] [INFO] [1749262352.587756189] [action_service]: action service execute move_right(90
, 1.5) succeeded
[action_service-4] [INFO] [1749262354.099185588] [action_service]: action service execute set_cmdvel(0,
0.5, 0, 1) succeeded
[action_service-4] [INFO] [1749262355.610367528] [action_service]: action service execute set_cmdvel(0,
-0.1, 0, 1) succeeded
[model_service-3] [INFO] [1749262359.233086694] [model_service]: "action": ['finishtask()'], "response":
我已经完成了所有的动作, 感觉就像完成了一场精彩的舞蹈表演! 如果你还有其他任务需要我帮忙, 随时告诉我哦!
```

Large Model Response Information Interpretation:

- **Decision Layer Large Model Output:** 1. Quickly move forward 1 meter, slowly move backward 0.5 meters, turn left 30 degrees, turn right 90 degrees, move left 0.5 meters, move right 0.1 meters. **Decision-Maintenance Model Output:** action: ['set_cmdvel(0.5, 0, 0, 2)', 'set_cmdvel(-0.1, 0, 0, 5)', 'move_left(30, 1.5)',

```
'move_right(90, 1.5)', 'set_cmdvel(0, 0.5, 0, 1)', 'set_cmdvel(0, -0.1, 0, 1)'],  
"response": Okay, I'll start executing your instructions now. First, I'll move  
forward quickly, then slowly backward, and then turn around, just like  
performing a complex dance. action: ['finishtask()'], "response": I've completed all the  
actions, and it feels like I've finished a wonderful dance performance! If you have any other  
tasks that you need my help with, feel free to let me know! The action list includes finish()
```



```
jetson@yahboom: ~  
[asr-5] [INFO] [1749265292.903460260] [asr]: 0  
[asr-5] [INFO] [1749265292.995522147] [asr]: 0  
[asr-5] [INFO] [1749265293.086186878] [asr]: 0  
[asr-5] [INFO] [1749265293.187068792] [asr]: 0  
[asr-5] [INFO] [1749265293.278757293] [asr]: 0  
[asr-5] [INFO] [1749265293.369376711] [asr]: 0  
[asr-5] [INFO] [1749265293.461203486] [asr]: 0  
[asr-5] [INFO] [1749265293.551661364] [asr]: 0  
[asr-5] [INFO] [1749265293.642409393] [asr]: 0  
[asr-5] [INFO] [1749265293.733644826] [asr]: 0  
[asr-5] [INFO] [1749265293.825658391] [asr]: 0  
[asr-5] [INFO] [1749265293.916962434] [asr]: 0  
[asr-5] [INFO] [1749265294.006844105] [asr]: 0  
[asr-5] [INFO] [1749265294.097273149] [asr]: 0  
[asr-5] [INFO] [1749265294.189699554] [asr]: 0  
[asr-5] [INFO] [1749265294.279358050] [asr]: 0  
[asr-5] [INFO] [1749265294.369934297] [asr]: 0  
[asr-5] [INFO] [1749265294.459665980] [asr]: 0  
[asr-5] [INFO] [1749265294.551012550] [asr]: 0  
[asr-5] [INFO] [1749265294.611719763] [asr]: 0  
[asr-5] [INFO] [1749265294.709735652] [asr]: 0  
[asr-5] [INFO] [1749265294.800441694] [asr]: 0  
[asr-5] [INFO] [1749265294.891975214] [asr]: 0  
[asr-5] [INFO] [1749265295.720371101] [asr]: 结束当前任务。  
[asr-5] [INFO] [1749265295.721262323] [asr]: okay😊, let me think for a moment...  
[model_service-3] [INFO] [1749265298.708199003] [model_service]: "action": ['finish_dialogue()'], "respo  
nse": 好的，任务已经结束了。有需要再叫我哦！  
[model_service-3] [INFO] [1749265304.175352702] [model_service]: The current instruction cycle has ended
```

3.2.2 Case 2

Similar to Case 1, after waking up and speaking, the robot performs operations according to the user's instructions.



```
[model_service-2] -----  
[model_service-2] -----  
[model_service-2] -----  
[model_service-2] -----  
[model_service-2] -----  
[model_service-2] -----  
[model_service-2] -----  
[model_service-2] -----  
[model_service-2] -----  
[model_service-2] -----  
[model_service-2] -----  
[model_service-2] -----  
[model_service-2] -----  
[model_service-2] -----  
[model_service-2] -----  
[model_service-2] -----  
[model_service-2] -----  
[model_service-2] [INFO] [1767700148.951756771] [ASR_Detect]: ASR Result: 请给我讲一个笑话。  
^[[2~[model_service-2] [INFO] [1767700157.592194611] [model_service]: "action": ['finish()'], "response": 当然可以！为什么电脑经常  
去健身房？因为它想要更快的处理器和更大的内存哦！
```

4. Code Analysis

This lesson applies the basic program framework of AI embodied intelligence. Code analysis refers to the **4. Embodied intelligent functions core source code** chapter.

5. Common Problem Solutions

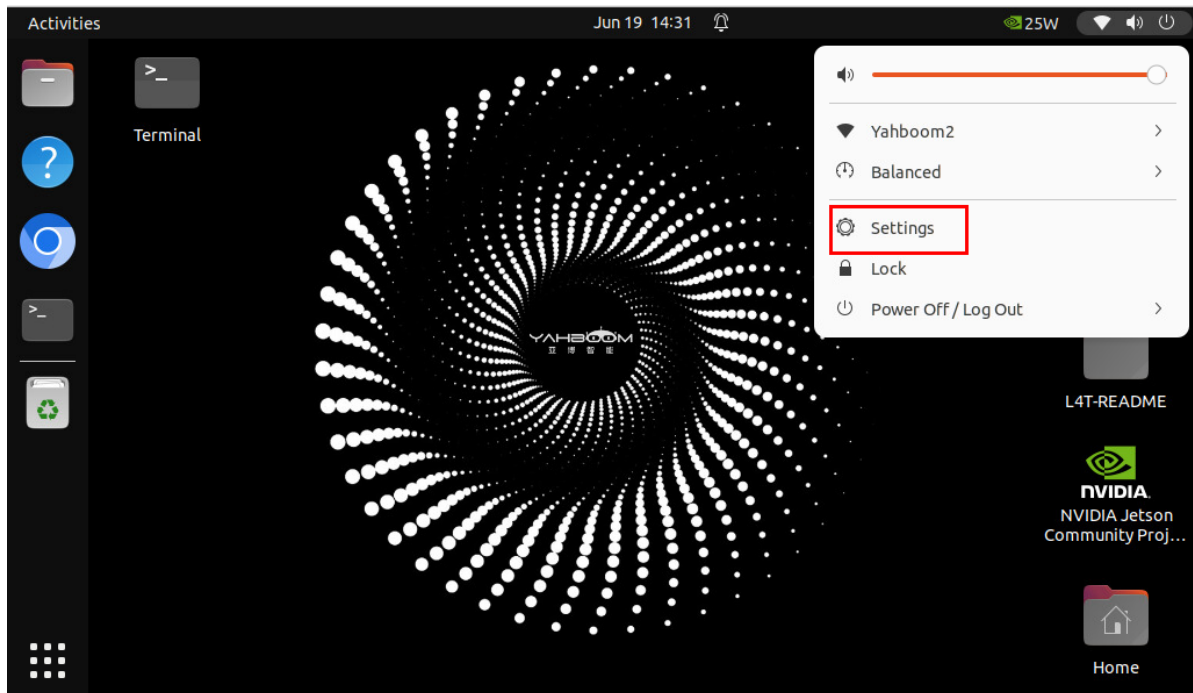
5.1 Microphone Recording Too Sensitive

If you find that the **VAD Voice Activity Detection** continuously displays "1-1-1-1" after the speaking phase ends, it indicates that the VAD detection is too sensitive. You can prioritize adjusting the microphone sensitivity; it's generally rare to adjust sensitivity at the software level.

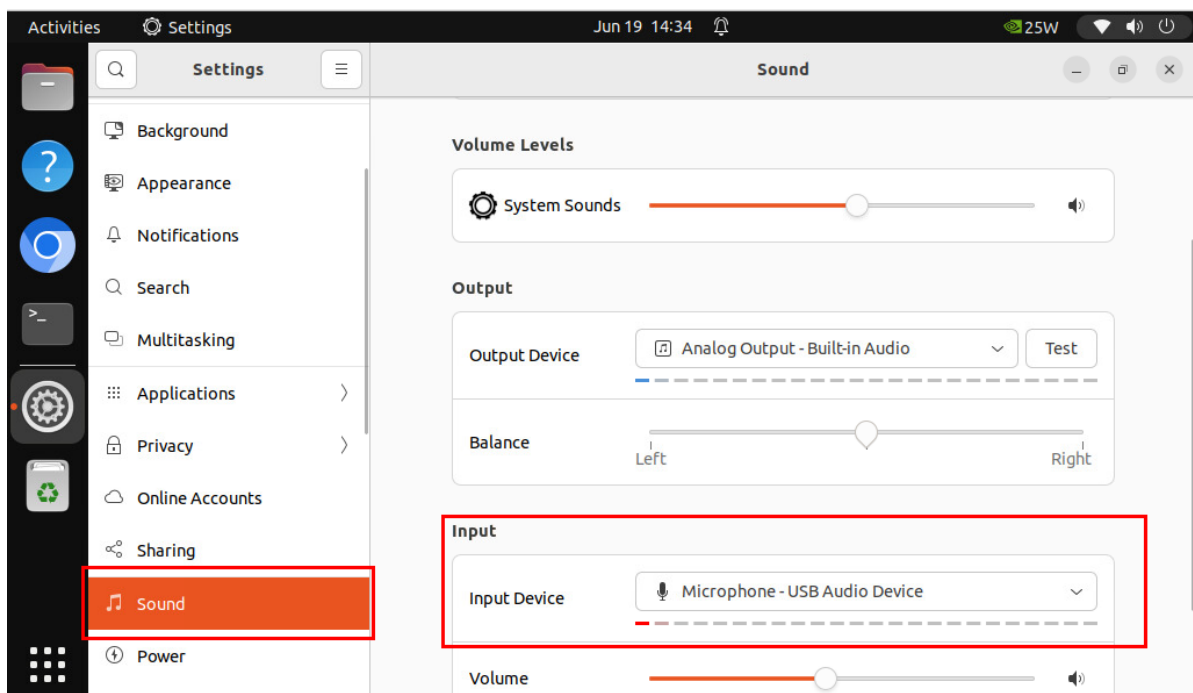
First, connect to the robot's onboard screen via VNC. Click the option bar in the upper right corner and find the **Settings** option.

[!NOTE]

- The operation method is similar for Raspberry Pi and Jetson Nano models; the microphone sensitivity is adjusted in the settings in the upper right corner, only the UI style differs.



Pull down the **Settings** list on the left, find the **Sound** option, and on the **Sound** page, find the **Input** audio input **Input Device**. Drag the **Volume** below to adjust the sensitivity. Try adjusting it to a suitable value while recording.



5.2 Microphone Recording Insensitive

If the speaker is far from the robot, the **VAD Voice Activity Detection** may not detect sound activity, causing the recording to end prematurely before the speaker finishes speaking. In this case, refer to the steps in **5.1 Microphone Recording Too Sensitive** to appropriately increase the microphone sensitivity.

Note:

- If the microphone sensitivity is set too high, it may increase the false positives for ambient noise, mistaking it for speech activity.

5.3 Software Adjustment of Speech Recognition Sensitivity

Configuration file path for the `multi_brains` package:

```
~/yahboomM3_ws/user_ws/src/m3_bringup/config/multi_brains/multi_brains_setting.yaml
```

- Edit the file using nano or another method:

```
nano
~/yahboomM3_ws/user_ws/src/m3_bringup/config/multi_brains/multi_brains_setting.yaml
```

- Locate the `VAD_MODE` parameter. The parameter range is 1-3; a higher number indicates higher sensitivity of the VAD in detecting speech activity.

```
#####
#ASR function setting
#语音识别功能设置
#####
USE_ONLINE_ASR : True                # whether to use online
ASR
ASR_SUPPLIER   : 'aliyun'            #ASR Supplier (only
effective when using online ASR): Chinese mainland: aliyun International: xunfei
ONLINE_ASR_MODEL : 'paraformer-realtime-v2'
ASR_THRESHOLD   : 3                 #ASR识别结果阈值,单位: 字符ASR
recognition result threshold. Unit: characters
WAKEUP_THRESHOLD : 5.0              #唤醒时间阈值,防止
WAKEUP_THRESHOLD时间内被多次唤醒,单位: 秒 wake-up time threshold, preventing
multiple awakenments within the WAKEUP_THRESHOLD time, unit: seconds
VAD_MODE: 1                        #vad灵敏度 VAD sensitivity
MAX_SILENCE_FRAMES: 90              #语音时长检测,单位: 帧数
Tail sound duration detection, unit: frames
```

5.4 End-of-Phonetic Detection Duration

If you need to adjust the end-of-phony detection duration, you can modify the `MAX_SILENCE_FRAMES` parameter value in the `multi_brains_setting.yaml` configuration file mentioned above.

```
jetson@yahboom: ~  
jetson@yahboom: ~ 122x24  
GNU nano 6.2 /home/jetson/yahboomM3_ws/user_ws/src/m3_bringup/config/multi_brains/multi_brains_setting.yaml  
#####  
USE_ONLINE_ASR : True #是否使用在线ASR  
ASR_SUPPLIER : 'aliyun' #ASR Provider (Only applicable when using online ASR): Mainland Chi  
ONLINE_ASR_MODEL : 'paraformer-realtime-v2' #ASR识别结果阈值, 低于该阈值则不进行后续处理, 单位: 字符  
ASR_THRESHOLD : 3 #唤醒时间阈值, 防止WAKEUP_THRESHOLD时间内被多次唤醒, 单位: 秒  
WAKEUP_THRESHOLD : 5.0 #vad灵敏度, 使用频率较少, 可以先调整麦克风灵敏度硬件灵敏度# VAD sen  
VAD_MODE: 2 #语音时长检测, 单位: 帧数, 1帧=30毫秒  
MAX_SILENCE_FRAMES: 90
```